Telco Customer Churn Analysis

Why customers leave a telecom company, and what the company can do about it

Project Report

Written in plain language for readers with no technical background.

1. Summary at a Glance

A telecom company sells phone and internet services to customers who pay every month. Some customers eventually cancel and leave. This is called **customer churn**. In this project, we studied the records of **7,043 customers** to understand who leaves and why.

> **The main result:** about **27 out of every 100 customers (26.54%)** left the company. Leaving was not random. Customers were far more likely to leave if they were on flexible month-to-month plans, were new to the company, used fiber optic internet, had no technical support or online security add-ons, or paid by electronic check.

The report explains what we did, what we found, and what the company can do to keep more of its customers.

2. Introduction

Imagine a gym that loses a quarter of its members every year. Each lost member means lost monthly income, and finding a new member usually costs much more than keeping an existing one. Telecom companies face the same problem on a much larger scale.

If a company knows **which customers are most likely to leave** and **why**, it can reach out to them early with better offers, better service or better plans, instead of finding out after they have gone.

2.1 Key terms used in this report

Term	Plain meaning
Churn	A customer cancelling the service and leaving the company.
Churn rate	The share of customers who left. A 26% churn rate means 26 out of every 100 customers left.
Tenure	How many months a customer has been with the company.
Contract	The plan type: month-to-month (cancel any time), one year, or two years.
Add-on services	Extra paid services such as Online Security, Tech Support, Online Backup and Device Protection.

3. Problem Statement

The company is losing a large share of its customers: about **26.54%** have churned. Every lost customer means lost monthly income, and replacing them costs more than keeping them. At the moment the company does not clearly know which types of customers are most likely to leave, or which factors are linked to their decision.

This project looks at customer details, account information and the services customers use, to find the factors most connected to churn, so the company can spot high-risk customers and act before they leave.

4. Objectives

The project had six goals:

- Find the overall share of customers who leave.
- Check whether personal details (gender, age group, having a partner or children) are linked to leaving.
- Check whether account details (contract type, length of stay, payment method) are linked to leaving.
- Check whether the services customers use (internet type, security, tech support, streaming and others) are linked to leaving.
- Identify the customer groups at the highest risk of leaving.
- Suggest practical, evidence-based actions to keep more customers.

5. About the Data

We used the well-known **Telco Customer Churn** dataset. It is like a large spreadsheet with **7,043 rows (one per customer)** and **21 columns** (facts about each customer). The most important column records whether the customer **left or stayed**.

Type of information	What it covers
About the person	Gender, whether they are a senior citizen, whether they have a partner, whether they have dependents (such as children).
About the account	Months with the company, contract type, payment method, paperless billing, monthly and total charges.
About the services	Phone service, multiple phone lines, internet type (DSL, fiber optic or none), online security, online backup, device protection, tech support, streaming TV and streaming movies.

6. What We Did (Our Approach)

The work followed four simple steps. The analysis was done in Python using a Jupyter Notebook.

Step	What happened
1. Look at the data	We checked how many customers and details the file holds, and looked for missing or odd values.
2. Clean the data	We fixed formatting problems and filled a small number of gaps. Some entries were blank because those customers had only just joined and had not been billed yet. We also changed the senior-citizen column from 0 and 1 to No and Yes so it is easier to read.
3. Compare groups	For each detail, we compared customers who left with customers who stayed. For example: do month-to-month customers leave more than customers on a two-year plan?
4. Explain the patterns	We turned the comparisons into findings and recommendations anyone can understand.

This type of work is called **exploratory data analysis**: exploring the data to discover patterns. It does not use complex prediction models.

7. What We Found

Overall: 26.54% of customers left. That is about 1 in 4. The table below shows the groups that stand out, comparing how many customers left in each group.

Factor	Group	Share who left
Contract	Month-to-month	about 43%
	One year	about 11%
	Two year	about 3%
Internet type	Fiber optic	about 42%
	DSL	about 19%
	No internet	about 7%
Tech Support	Without it	about 42%
	With it	about 15%
Online Security	Without it	about 42%
	With it	about 15%
Payment method	Electronic check	about 45%
	Mailed check, bank transfer or credit card	about 15% to 19%
Age group	Senior citizens	about 42%
	Non-seniors	about 24%
Partner	No partner	about 33%
	Has a partner	about 20%
Dependents	No dependents	about 31%
	Has dependents	about 16%

The contract, senior citizen and gender figures are calculated exactly from the data. The other figures are rounded estimates, so treat them as approximate.

7.1 The main reasons customers leave

1. Flexible contracts make it easy to leave.

Customers on month-to-month plans are by far the most likely to leave. They make up roughly three quarters of all customers who left. Customers on one-year and especially two-year contracts almost always stay.

2. New customers are at the highest risk.

Most departures happen in the first months of service. Customers who have stayed for a long time rarely leave.

3. Fiber optic customers leave more often.

About 42% of fiber optic customers left, compared with about 19% of DSL customers. This may point to price, service quality or competition, but the data alone cannot say which.

4. Customers without support and security services leave more.

Customers who do not have Tech Support or Online Security leave at around 42%, compared with around 15% for those who do. Online Backup and Device Protection show the same pattern. These services may help customers feel looked after.

5. Paying by electronic check is a warning sign.

Electronic check users leave at around 45%, far more than customers who pay by mailed check, bank transfer or credit card. Automatic payments may keep customers more settled in.

6. Some customer profiles leave more.

Senior citizens, customers without a partner and customers without dependents are more likely to leave.

7.2 What does NOT seem to matter

Gender makes no real difference: about 27% of women and about 26% of men left. Having **phone service**, having **multiple phone lines**, and using **streaming TV or movies** also showed little or no difference in who left.

7.3 Two points that are easy to misread

- **Senior citizens:** there are far fewer senior customers than non-senior customers, so more non-seniors left in total. But seniors are much more likely to leave *as a group* (about 42% versus 24%).
- **Phone service:** most customers have phone service, so many leavers have it too. That does not mean phone service causes people to leave.

8. Recommendations

Based on the findings, the company could take these practical steps:

Action	Why
Offer discounts or perks to move month-to-month customers onto one or two-year plans.	Longer contracts have by far the lowest departure rates.
Give new customers extra attention in their first 3 to 6 months (welcome calls, check-ins, small offers).	This is when most customers decide to leave.
Bundle or discount Tech Support and Online Security, especially with fiber optic plans.	Customers who have these services leave far less.
Review fiber optic pricing and service quality.	Fiber customers leave at more than twice the rate of DSL customers.
Encourage automatic payments with a small incentive.	Electronic check users leave much more often.
Create targeted retention offers for senior citizens and customers living alone.	These groups leave more often than others.

9. Limitations and Next Steps

9.1 Things to keep in mind

- **Patterns, not proof.** We found that certain things go together with leaving. This does not prove that one thing *causes* the other.
- **No statistical testing.** We compared group percentages but did not run formal tests to confirm each difference is statistically significant.
- **Some topics were not studied in depth,** including monthly charges, total charges and paperless billing.

9.2 Possible next steps

- Study how monthly and total charges relate to leaving.
- Build a prediction model that gives each customer a score showing how likely they are to leave, so the company can contact the riskiest customers first.
- Estimate how much income the company loses to churn, and how much it could save by cutting the churn rate by a few points.

10. Conclusion

Customers do not leave at random. Most of the risk sits with **new customers on flexible month-to-month plans, using fiber optic internet, without support or security add-ons, and paying by electronic check**. By offering longer-term plans, supporting new customers early, promoting support services and encouraging automatic payments, the company has clear, practical ways to reduce churn and protect its income.